



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Data on the Falcon 9 rockets were collected from a SpaceX REST API and scraped from the SpaceX wikipedia Page.
- Visualizations of the data were made using the seaborn libraries and SQL queries.
- Tuned four classifiers (Logistic Regression, SVM, Decision Tree, KNN) with GridSearchCV (cv=10) after standardizing features
- Overall first-stage landing success rate in the dataset is 66.7%; best models reach ~83% accuracy on held-out test data, with Decision Tree scoring highest on cross-validation (88.9%)
- Flight number, orbit type, and payload mass are the clearest drivers of landing success

Introduction

- SpaceX advertises Falcon 9 launches at \$62M, versus \$165M+ from other providers — largely because SpaceX reuses the first stage
- The true cost of launches can most accurately predict where the rocket will land.
- A competitor could use this to inform a competitive bid against SpaceX
- Goal: build a machine-learning pipeline that predicts Falcon 9 first-stage landing success from launch data

Section 1

Methodology

Methodology

Data collection methodology:

SpaceX REST API (api.spacexdata.com) requests, plus web scraping of the Wikipedia Falcon 9/Heavy launch table with BeautifulSoup

Perform data wrangling

Engineered a binary Class label from the landing Outcome column; checked missing values (LandingPad ~29% missing, expected for non-landing attempts)

Perform exploratory data analysis (EDA) using visualization and SQL

Perform interactive visual analytics using Folium and Plotly Dash

Perform predictive analysis using classification models

Standardized features, 80/20 train-test split, GridSearchCV (cv=10) over Logistic Regression, SVM, Decision Tree, and KNN; compared test-set accuracy

Data Collection

- Two data sources combined: the SpaceX REST API (structured launch/rocket/core records) and a Wikipedia table (via web scraping) used as a cross-check
- Flow: request data -> normalize/flatten into a DataFrame -> filter to Falcon 9 -> handle missing values -> export CSV for downstream labs

Data Collection – SpaceX API

- Requested past launch records from the SpaceX REST API (api.spacexdata.com)
- Normalized nested JSON fields (rocket, payloads, launchpad, cores) into flat columns

Request GET api.spacexdata.com

-> Flatten nested JSON (rocket/payload/core) into columns

-> Filter to Falcon 9 launches

-> Handle missing values

-> Export dataset_part_1.csv

- Sent an HTTP GET request (with a browser User-Agent header) to a fixed Wikipedia snapshot: "List of Falcon 9 and Falcon Heavy launches" (9 June 2021 revision)
- Parsed the HTML with BeautifulSoup and extracted the launch-records table; cleaned header names with a custom `extract_column_from_header()` helper (stripped `<br>`, `<a>`, `<sup>` tags)
- Iterated each table row with helper functions to pull Flight No., Date, Time, Booster Version, Launch Site, Payload, Payload Mass, Orbit, Customer, Launch Outcome, and Booster Landing status into a dict, then built a DataFrame and exported `spacex_web_scraped.csv`

<https://github.com/Taylor-jat/IBM-data-scientist-capstone/blob/master/jupyter-labs-webscraping.ipynb>

GET Wikipedia page (static snapshot)

- > Parse HTML with BeautifulSoup
- > Extract table headers & rows
- > Clean & assemble `launch_dict`
- > Export `spacex_web_scraped.csv`

Data Wrangling

- Loaded the 90-record launch dataset; checked missing values (LandingPad ~28.9% missing, expected since ocean/no-landing attempts have no pad) and column dtypes
- Counted launches per site (CCAFS SLC 40: 55, KSC LC 39A: 22, VAFB SLC 4E: 13) and per orbit (GTO 27, ISS 21, VLEO 14, PO 9, LEO 7, SSO 5, MEO 3, ES-L1/HEO/SO/GEO 1 each)
- Defined a "bad outcome" set {False ASDS, False Ocean, False RTLS, None ASDS, None None} representing unsuccessful landings
- Engineered the binary Class label (1 = landed, 0 = did not land) from the Outcome column; overall success rate = 66.7%
- Exported the labeled dataset (dataset_part_2.csv) for downstream EDA and modeling

<https://github.com/Taylor-jat/IBM-data-scientist-capstone/blob/master/labs-jupyter-spacex-Data%20wrangling.ipynb>

- Used seaborn catplot/scatterplot/barplot/lineplot to explore what drives landing success: FlightNumber vs PayloadMass, FlightNumber vs LaunchSite, PayloadMass vs LaunchSite, success rate by Orbit, FlightNumber vs Orbit, PayloadMass vs Orbit, and yearly success-rate trend
- Key findings: success becomes more common at higher flight numbers; VAFB SLC 4E never launched payloads over ~10,000 kg; LEO success correlates with flight number while GTO does not; success rate rose steadily from 2013 to 2020
- Selected 12 features for modeling and one-hot encoded the categorical ones (Orbit, LaunchSite, Serial, LandingPad) via `pd.get_dummies`, producing an 83-column, 90-row feature set (`dataset_part_3.csv`)

<https://github.com/Taylor-jat/IBM-data-scientist-capstone/blob/master/edadataviz.ipynb>

- Unique launch sites: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E
- Total payload mass carried for NASA-related customers: 107,010 kg
- Average payload mass for booster version F9 v1.1: 2,534.67 kg
- Total mission outcomes by landing type: Success 38, Success (drone ship) 14, Success (ground pad) 9, No attempt 22, Failure (drone ship) 5, Controlled (ocean) 5, Failure 3, Uncontrolled (ocean) 2, Failure (parachute) 2, Precluded (drone ship) 1
- Booster versions that carried the maximum payload mass: 12 F9 B5 boosters (e.g., B1048.4, B1049.4, B1051.3, B1056.4, B1060.2, B1058.3...)
- Only two failed drone-ship landings recorded in 2015: F9 v1.1 B1012 (Jan, CCAFS LC-40) and F9 v1.1 B1015 (Apr, CCAFS LC-40)

https://github.com/Taylor-jat/IBM-data-scientist-capstone/blob/master/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Build an Interactive Map with Folium

- Circle markers + text labels for each of the 4 launch sites, plotted on a map centered near Houston, TX
- A MarkerCluster of every individual launch record, color-coded green (success) / red (failure), to spot which sites have higher success rates at a glance
- A MousePosition readout plus PolyLines and distance calculations (haversine formula) from a launch site to its nearest coastline, city, railway, and highway
- Findings: all launch sites sit close to the coast (for downrange/ocean recovery) and near the equator (rotational boost for GTO missions), while keeping distance from nearby cities

https://github.com/Taylor-jat/IBM-data-scientist-capstone/blob/master/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

- Dropdown to pick a launch site (or "All Sites"), driving both charts below
- Pie chart: total successful launches by site (All Sites), or the success/fail split for one selected site
- RangeSlider (bound to the data's actual min/max payload) plus a scatter plot of Payload Mass vs. Class, colored by Booster Version Category, filtered by both the selected site and the payload range
- Added these interactions so a reviewer can compare sites at a glance, then drill into any one site's success rate and see how payload/booster generation relate to landing outcome

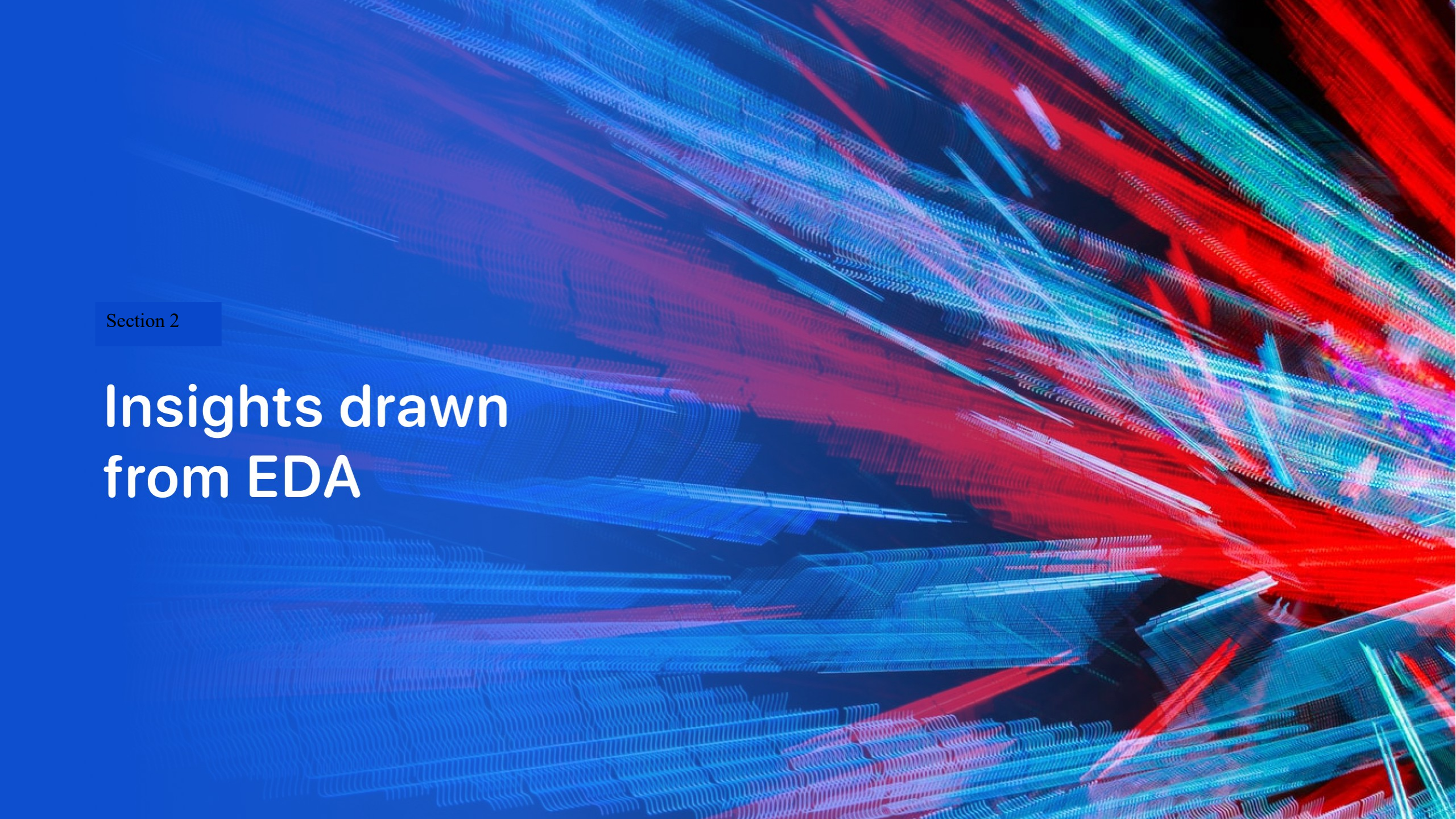
<https://github.com/Taylor-jat/IBM-data-scientist-capstone/blob/master/spacex-dash-app.py>

Predictive Analysis (Classification)

- Standardized all features (StandardScaler) and split into 72 training / 18 test samples (80/20, random_state=2)
- Tuned 4 classifiers with GridSearchCV (cv=10): Logistic Regression (C, penalty, solver), SVM (kernel, C, gamma), Decision Tree (criterion, splitter, max_depth, max_features, min_samples_leaf/split), and KNN (n_neighbors, algorithm, p)
- Best cross-validation accuracy: Decision Tree 88.9%, KNN 84.8%, SVM 84.8%, Logistic Regression 84.6%
- Test-set accuracy (18 held-out samples): Logistic Regression, SVM, and Decision Tree all scored 83.3%
- [Add GitHub URL to your notebook here]

Results

- Exploratory data analysis results
 - Success rate rose from 2013-2020; flight number, orbit type, and payload mass are the clearest predictors of landing success
- Interactive analytics demo in screenshots
 - Folium map shows sites clustered near the coast/equator, with visibly higher success rates at some sites than others
- Predictive analysis results
 - Best models reach ~83% test accuracy; Decision Tree has the highest cross-validated accuracy (88.9%)

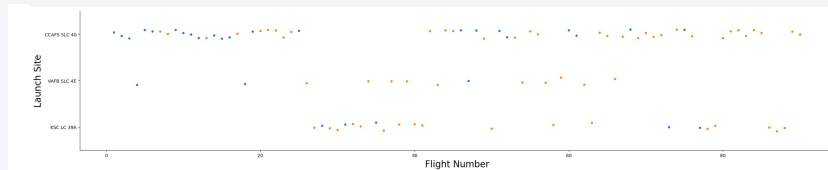


Section 2

Insights drawn from EDA

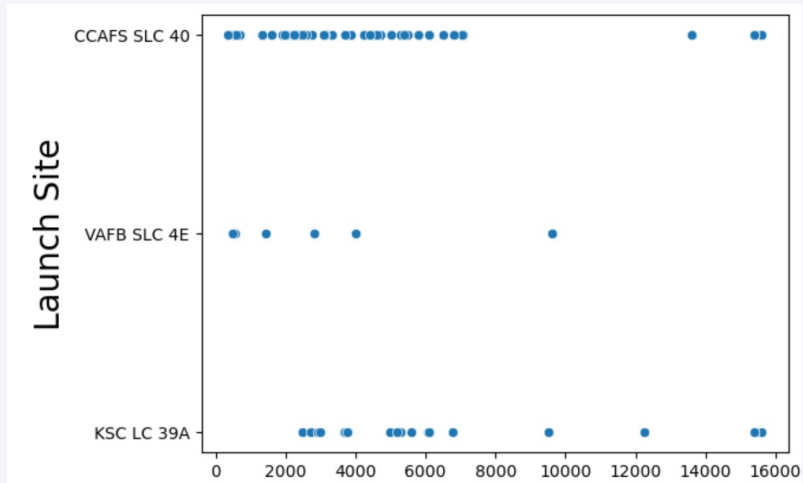
Flight Number vs. Launch Site

- Later flight numbers show more successful landings across all three sites; early flights (low Flight Number) skew toward failed landings
- The following graph shows a high concentration of successful launches from KSC LC-39A



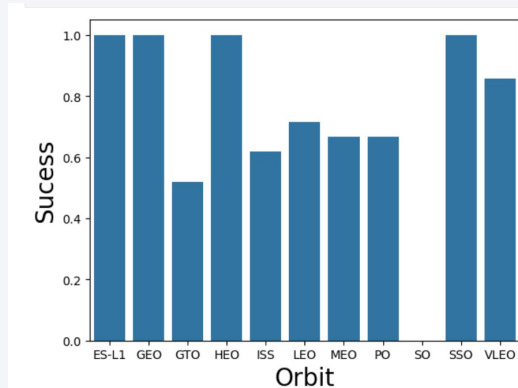
Payload vs. Launch Site

- There were no launches from VAFB SLC-4E that had a payload over 10,000 kg
- CCAFS SLC-40 and KSC LC-39A both handle the full range of payload masses, including the heaviest payloads



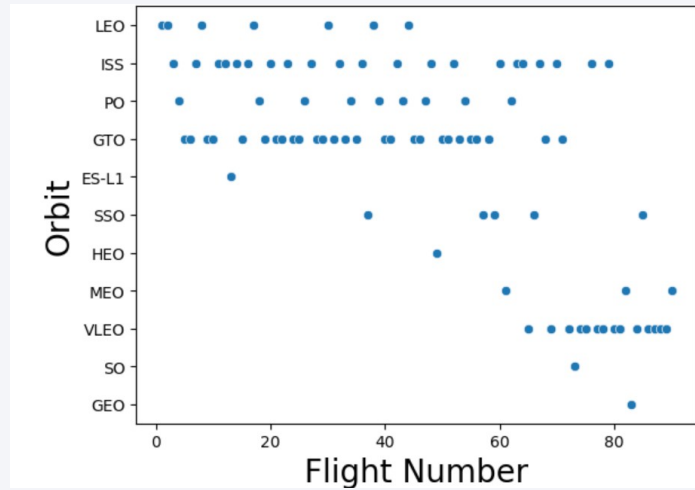
Success Rate vs. Orbit Type

- Success rate by orbit: ES-L1, GEO, HEO, and SSO all achieved 100%; VLEO 86%, LEO 71%, MEO and PO 67%, ISS 62%, GTO 52%, and SO 0% (single flight)
- GTO shows the lowest multi-flight success rate, consistent with its higher share of older/expendable missions



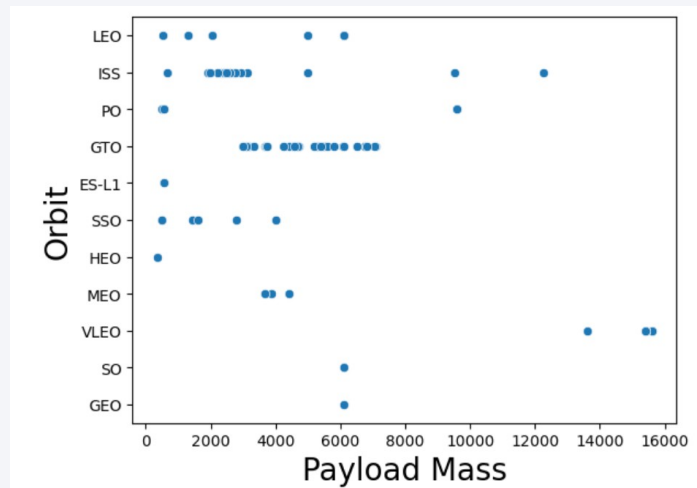
Flight Number vs. Orbit Type

- In LEO, success appears correlated with flight number - later flights land more often
- In GTO, there is no clear relationship between flight number and landing success



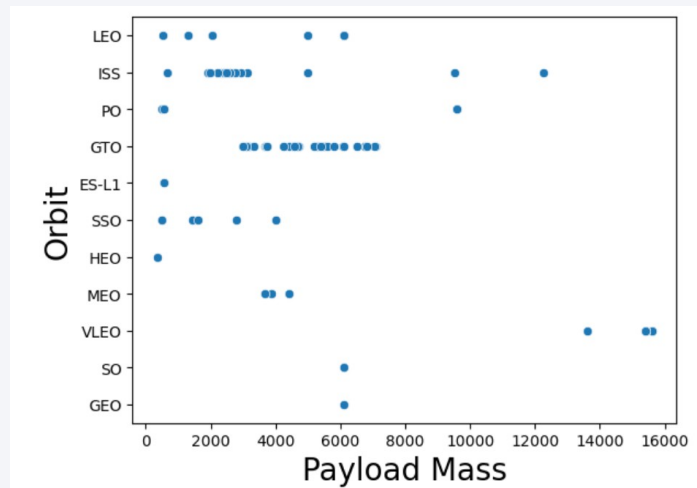
Payload vs. Orbit Type

- Heavy payloads land successfully more often in Polar, LEO, and ISS orbits
- In GTO, successful and unsuccessful landings both occur across the payload range, making payload mass a weaker predictor there



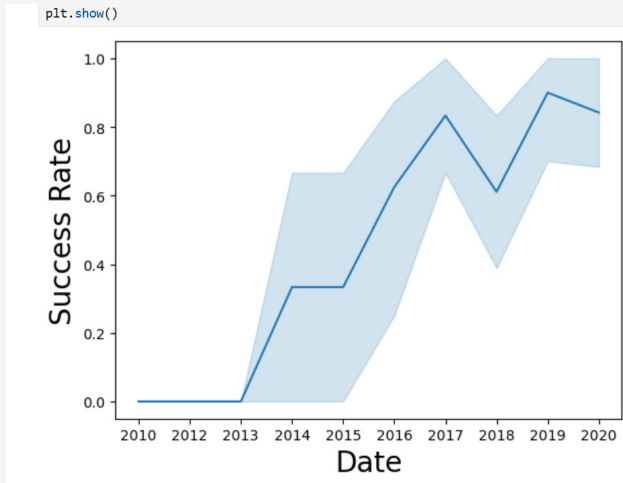
Payload vs. Orbit Type

- Heavy payloads are shown to be a fair predictor in Polar, LEO, and ISS orbits
- The data shows that payload mass is a weaker predictor there.



Launch Success Yearly Trend

- Success rate rose steadily from 2013 through 2020, reflecting SpaceX's growing landing expertise and technology maturity
- The shaded confidence band narrows over time as launch volume increased, indicating more stable, predictable outcomes



All Launch Site Names

- CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E
- Query: `SELECT DISTINCT(Launch_Site) FROM SPACEXTABLE`

Launch Site Names Begin with 'CCA'

- First 5 records where Launch_Site LIKE "%CCA%" — all from CCAFS LC-40, spanning the earliest Dragon/NASA CRS missions (2010-2013)
- Query: `SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE "%CCA%" LIMIT 5`

Total Payload Mass

- Total payload mass carried by boosters for NASA-related customers: 107,010 kg
- Query: `SELECT SUM(PAYLOAD_MASS_KG_) FROM SPACEXTABLE WHERE Customer LIKE "%NASA%"`

Average Payload Mass by F9 v1.1

- Average payload mass carried by booster version F9 v1.1: 2,534.67 kg
- Query: `SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Booster_Version LIKE "%F9 v1.1%"`

First Successful Ground Landing Date

- Earliest record in the table is 2010-06-04 (first Falcon 9 flight)
- Note: to get the first *ground-pad* success specifically, filter on Landing_Outcome = 'Success (ground pad)' before taking MIN(Date) — the query as run only finds the overall earliest row, so double-check this result against that corrected query

Successful Drone Ship Landing with Payload between 4000 and 6000

- Boosters that succeeded on a drone ship with payload between 4,000-6,000 kg: F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Success: 38, Success (drone ship): 14, Success (ground pad): 9, No attempt: 21 (+1 with trailing space), Failure (drone ship): 5, Controlled (ocean): 5, Failure: 3, Uncontrolled (ocean): 2, Failure (parachute): 2, Precluded (drone ship): 1

Boosters Carried Maximum Payload

- 12 F9 B5 boosters tied for the maximum payload mass, including B1048.4, B1049.4, B1051.3, B1056.4, B1048.5, B1051.4, B1049.5, B1060.2, B1058.3, B1051.6, B1060.3, B1049.7

2015 Launch Records

- Only 2 failed drone-ship landings recorded in 2015: January - F9 v1.1 B1012 (CCAFS LC-40); April - F9 v1.1 B1015 (CCAFS LC-40)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Ranked landing-outcome counts between 2010-06-04 and 2017-03-20: No attempt (10), Success (drone ship) & Failure (drone ship) tied (5 each), Success (ground pad) & Controlled (ocean) tied (3 each), Uncontrolled (ocean) & Failure (parachute) tied (2 each), Precluded (drone ship) (1)

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite image of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible as a thin, curved line separating the dark surface from the deep blue of space.

Section 3

Launch Sites Proximities Analysis

All Launch Sites on a Global Map

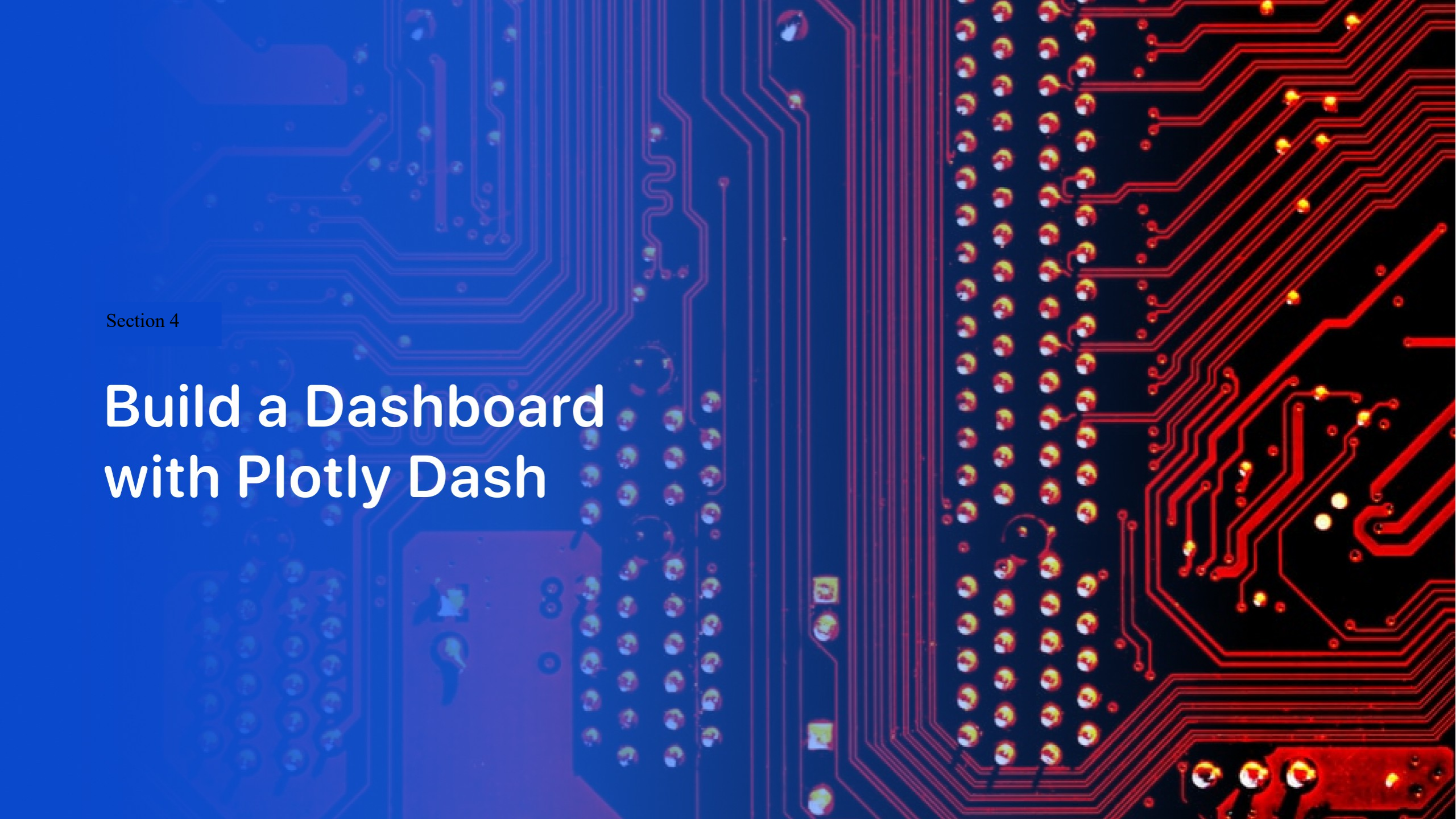
- [Insert screenshot from your notebook here]
- Circle markers with labels for CCAFS LC-40/SLC-40, KSC LC-39A, and VAFB SLC-4E; all cluster near the coast, close to the equator

Color-Labeled Launch Outcomes

- [Insert screenshot from your notebook here]
- Green markers = successful landings, red = failed, clustered per site; visually confirms which sites have higher success ratios

Launch Site Proximity to Coastline

- [Insert screenshot from your notebook here]
- PolyLines from CCAFS LC-40 to its nearest coastline point (~0.86 km), plus lines to the nearest city, railway, and highway with distances labeled
- Sites sit very close to the coast for ocean recovery/downrange safety, while keeping a buffer distance from nearby cities

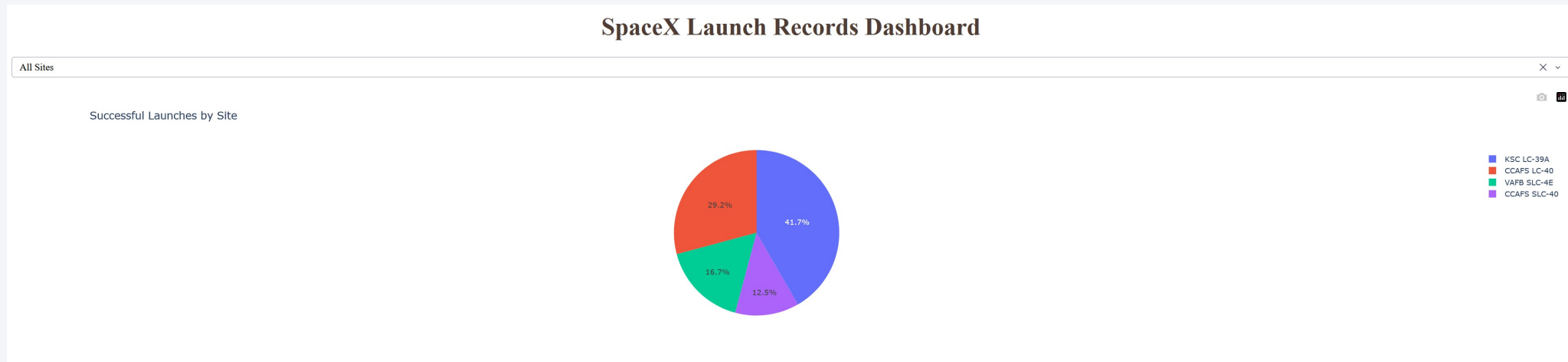


Section 4

Build a Dashboard with Plotly Dash

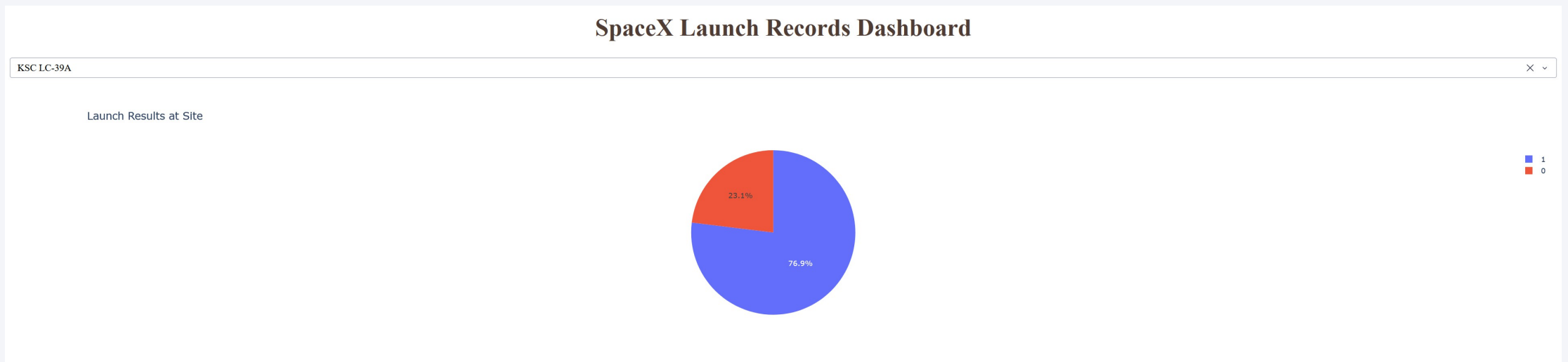
Successful Launches by Site (All Sites)

- KSC LC-39A accounts for the largest share of successful launches (41.7%), followed by CCAFS LC-40 (29.2%), VAFB SLC-4E (16.7%), and CCAFS SLC-40 (12.5%)
- [Add GitHub URL to your notebook here]



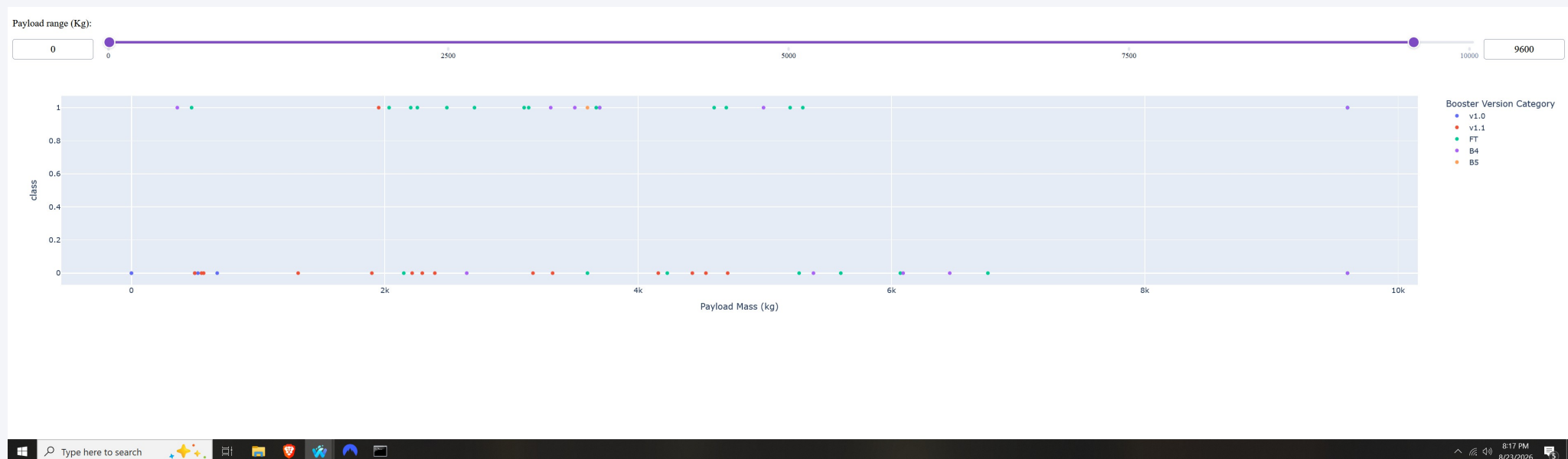
Highest Success-Ratio Site: KSC LC-39A

- KSC LC-39A has the best per-site success ratio: 76.9% of its launches succeeded vs. 23.1% that did not
- [Add GitHub URL to your notebook here]



Payload vs. Launch Outcome (All Sites)

- Across the full payload range (0-9,600 kg) and all sites, newer booster versions (FT, B4, B5) cluster among the successes, while older versions (v1.0, v1.1) skew toward failures at low payloads
- Success is fairly consistent from 0-6,000 kg; very few launches were attempted above 6,000 kg
- [Add GitHub URL to your notebook here]



Section 5

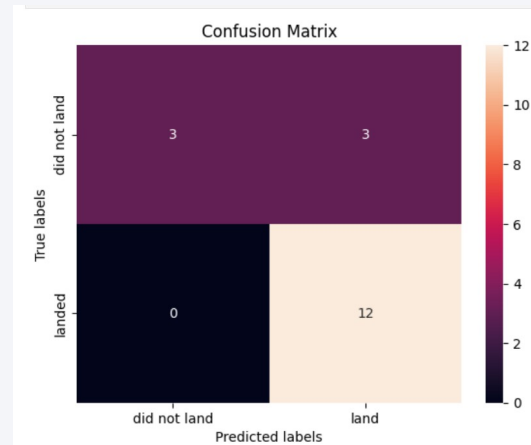
Predictive Analysis (Classification)

Classification Accuracy

- [Insert screenshot from your notebook here] (bar chart of best_score_/test accuracy per model)
- Cross-validation accuracy: Decision Tree 88.9%, KNN 84.8%, SVM 84.8%, Logistic Regression 84.6%
- Test-set accuracy: all four models tied at 83.3% (Logistic Regression, SVM, Decision Tree, and KNN, using `knn_cv.score(X_test, Y_test)`)
- Decision Tree has the highest cross-validated accuracy overall and is the recommended model

Confusion Matrix

- Confusion matrix shown is for the Decision Tree model (highest cross-validated accuracy, 88.9%; test accuracy 83.3%)
- Did not land: 3 correctly predicted, 3 misclassified as landed. Landed: 12 correctly predicted, 0 misclassified as did-not-land
- The model's only error mode is over-predicting successful landings; it never misses an actual landing



Conclusions

- Falcon 9 first-stage landing success has improved substantially since 2013, and can be predicted with ~83% test accuracy
- Flight number, orbit type, and payload mass are the strongest predictors of landing success
- Decision Tree achieved the highest cross-validated accuracy (88.9%) among the four models tested
- Launch sites are located close to the coast and near the equator, consistent with recovery logistics and orbital-mechanics advantages
- Reliable landing prediction lets a competitor estimate SpaceX's true launch cost and price a competitive bid accordingly

Appendix

- Include: SpaceX API notebook, web-scraping notebook, data-wrangling notebook, EDA (viz + SQL) notebooks, Folium map notebook, Plotly Dash app, predictive-analysis notebook
- Link each notebook's GitHub URL here for peer review

Thank you!

